

SYSTEM DESIGN

4.1 Basic Modules

The **Smart Event Management and Ticket Booking System** is divided into six functional modules. Each module performs a specific task and together they provide complete event management functionality.

Module 1: User Management

This module manages user registration, login, logout, and profile management. It authenticates users and allows them to update their personal information securely.

Functions:

- User Registration
 - User Login
 - User Logout
 - Profile Management
-

Module 2: Event Management

This module allows administrators to create, update, delete, and manage events. Users can browse available events and view detailed event information.

Functions:

- Create Event
 - Update Event
 - Delete Event
 - View Events
-

Module 3: Booking Management

This module handles ticket booking operations. Users can book tickets, view booking history, and administrators can manage booking records.

Functions:

- Book Ticket
 - View Booking
 - Cancel Booking
 - Manage Bookings
-

Module 4: Payment Management

This module processes ticket payments and stores payment details for each booking.

Functions:

- Process Payment
 - View Payment Status
 - Update Payment Status
-

Module 5: Feedback Management

This module enables users to submit ratings and feedback after attending an event. Administrators can review all submitted feedback.

Functions:

- Submit Feedback
 - View Feedback
 - Manage Feedback
-

Module 6: Administration Module

This module provides complete administrative control over the application.

Functions:

- Manage Users
 - Manage Events
 - Manage Bookings
 - View Payments
 - Generate Reports
-

4.2 Data Design

The Smart Event Management and Ticket Booking System uses **MongoDB Atlas** as its database. Data is stored in separate collections to maintain consistency and improve performance.

The database stores information related to users, events, bookings, payments, and feedback. Each collection contains documents that represent real-world entities. The collections are related through unique identifiers, allowing efficient retrieval and management of records.

The major collections used in the system are:

- User
- Event
- Booking
- Payment
- Feedback

4.2.1 Schema Design (Data Dictionary)

USER Collection

Field Name	Data Type	Description
userId	Integer	Unique User ID
name	String	User Name
email	String	Email Address
password	String	User Password
phone	String	Mobile Number
role	String	User Role (Admin/User)

EVENT Collection

Field Name	Data Type	Description
eventId	Integer	Unique Event ID
title	String	Event Title
description	String	Event Description
category	String	Event Category
organizer	String	Organizer Name
venue	String	Event Venue
date	Date	Event Date
time	String	Event Time
capacity	Integer	Maximum Capacity
price	Double	Ticket Price

BOOKING Collection

Field Name	Data Type	Description
bookingId	Integer	Unique Booking ID
userId	Integer	User Reference
eventId	Integer	Event Reference
bookingDate	Date	Booking Date
ticketQuantity	Integer	Number of Tickets
status	String	Booking Status

PAYMENT Collection

Field Name	Data Type	Description
paymentId	Integer	Unique Payment ID
bookingId	Integer	Booking Reference
amount	Double	Payment Amount
paymentMethod	String	Payment Method
paymentStatus	String	Payment Status
paymentDate	Date	Payment Date

FEEDBACK Collection

Field Name	Data Type	Description
feedbackId	Integer	Unique Feedback ID
userId	Integer	User Reference
eventId	Integer	Event Reference
rating	Integer	User Rating
comment	String	Feedback Comment
feedbackDate	Date	Date of Feedback